

EFFICIENT MULTIMODAL INFERENCE · EDGE + MOBILE

SmolVLM: Redefining **Small** and **Efficient** Multimodal Models

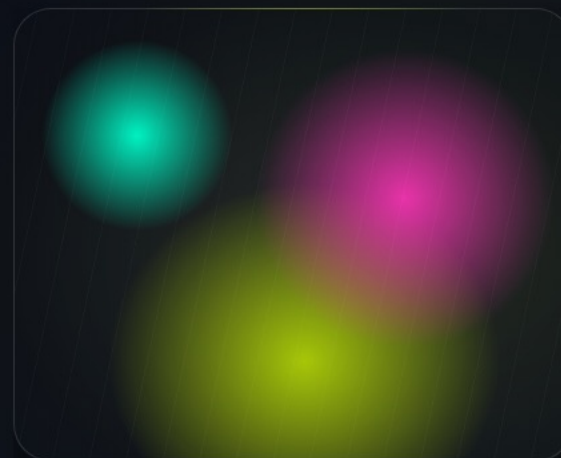
A family of compact vision-language models enabling sub-1GB inference via compute allocation, pixel-shuffle token compression, and context scaling.

Hugging Face

Stanford University

arXiv:2504.05299

Space Navigate
Target: ML researchers · on-device AI · grad students



PROBLEM FRAMING

Large VLMs Are Powerful but Impractical for Edge Deployment

Why edge VLMs are hard

- Large VLMs require significant resources (e.g., memory, compute power)
- Edge devices have limited resources (e.g., memory, compute power)
- Large VLMs often require high-quality data for training
- Edge devices often have limited data availability

ARCHITECTURE

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone

Image Splitting

Pixel Shuffle

FINDING 1

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Result (small LM regime)

- `ViT-S/16` + `LLaMA-2-7B` beats `ViT-L/16` + `LLaMA-2-7B`
- `ViT-S/16` + `LLaMA-2-7B` beats `ViT-L/16` + `LLaMA-2-7B` (smaller encoder, smaller LM)

Aggressive Pixel Shuffle ($r=4$) Cuts Visual Tokens 16× for Small Models

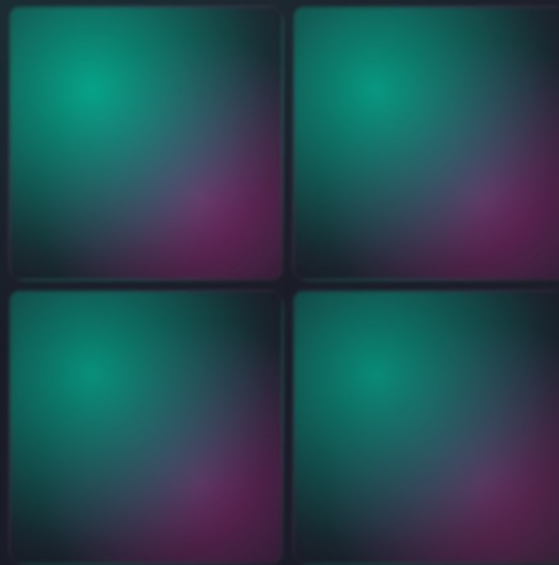
Mechanism: space $\rightarrow$ depth

BEFORE (MORE TOKENS)



Spatial grid $\rightarrow$ long visual token sequence

AFTER (FEWER TOKENS)



Tokens $\div r^2$, channels $\times r^2$

With pixel shuffle factor $r=4$

16× fewer visual tokens

Reduces attention overhead and helps long-context modeling in small VLMs.

FINDING 2

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

What changed

- Context length extended from 2K → 16K tokens for significant performance gains.
- Increase RoPE base from 10K to 273K.
- Fine-tune on mixed long/short context data.

Deployment choices

- SmolVLM-2.2B uses 16K context.
- SmolVLM-256M / 500M use 8K context.

Why it matters

- Longer context supports higher image resolution (more visual tokens) without truncation.
- Pairs naturally with token compression to keep attention manageable.

Context extension recipe

RoPE base 10K → 273K

Then fine-tune with mixed long/short contexts.

BENCHMARKS (13 TASKS)

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Structured comparison

MODEL	PARAMS	AVG (ALL)	RAM (B~1)
SmolVLM 256M	256M	44.0%	0.8 GB
SmolVLM 500M	500M	51.0%	1.2 GB
SmolVLM 2.2B	2.2B	59.8%	4.9 GB
Best Efficient OS Molmo-A1B-7B	7B	—	27.7 GB
Best Efficient OS InternVL2-2B	2B	—	—

Avg = mean across 13 benchmarks. Efficient OS shown per-benchmark in the paper; RAM comparison provided for Molmo-A1B-7B.

Highlights from the paper

- SmolVLM evaluated on 13 benchmarks spanning single-image, multi-task, and video.
- SmolVLM-256M runs with 0.8 GB GPU RAM and outperforms the 300× larger Idefics 80B.
- SmolVLM-2.2B reaches 59.8% average while using 4.9 GB RAM.

Takeaway

Efficiency beats raw scale

Architectural choices can close the gap to much larger models under tight memory.

Real-Time Inference on iPhones: Up to ~80 Tokens/Second on Apple Silicon

 HuggingSnap app and throughput figure

Figure: HuggingSnap + throughput across Apple Silicon devices

Reported throughput

- SmolVLM runs locally via the [HuggingSnap](#) iOS app (no cloud dependency).
- SmolVLM-256M achieves up to **~80 tokens/sec** on Pro M1 Max 64GB.
- At batch size 64, the 256M model reaches nearly **1000 tokens/sec** on high-end devices.
- Even on a MacBook Air M3 16GB: **~55 tokens/sec** (256M model).
- Throughput scales roughly **linearly** with batch size for the smallest model.

Numbers are taken from the paper's on-device benchmarks summary.

Key Takeaways: Design Principles for Efficient Multimodal Models

1) Balance compute

For compact LMs, smaller encoders can outperform larger ones.

93M SigLIP-B/16 > 428M SigLIP-S0400M (at 135M LM)

2) Compress tokens

Use aggressive pixel shuffle to cut attention cost.

$r=4 \Rightarrow 16\times$ fewer visual tokens (small models)

3) Scale context

Extend context length with RoPE base adjustment + mixed-context tuning.

2K $\rightarrow$ 16K; RoPE base 10K $\rightarrow$ 273K

4) Target tight RAM budgets

- 0.8 GB RAM inference is achievable (SmoVLM-256M).
- Competitive averages up to 59.8% (SmoVLM-2.2B).

5) Reproducible + open

- All weights, datasets, code, and a mobile app are **fully open-sourced**.
- Enables follow-up research on efficient multimodal training + deployment.

Closing thought

Design the bottleneck first.

In edge VLMs, the bottleneck is memory + tokens — not parameters alone.